Cooperative Wireless Communications: Present and Future

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Thesis Presentation for
Programa de Doutoramento en Enxeñería Telemática

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Motivation

> Mobile data traffic will increase nearly 11-fold between 2013 and 2018
>
> **Cisco Visual Networking Index**

> The wireless industry has reached the theoretical limit of how fast networks can go
>
> **K. Fitcher, Connected Planet**

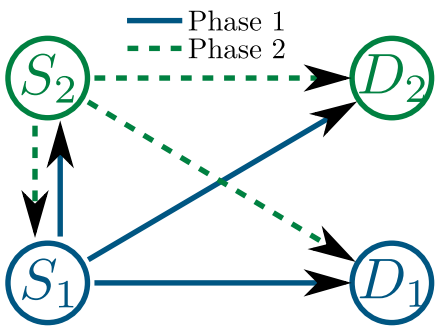
One of these must be wrong

We know

1. The Shannon limit on a (AWGN) link
2. How to achieve it

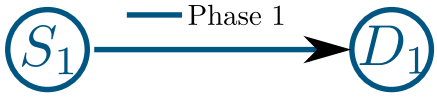
But

1. The capacity of this



2-User Cooperative Channel

2. Is not $2\times$ the capacity of one of these



Point to Point Channel

So...

1. There is still room to improve rates
2. But analyzing one link at a time does not help
3. We must improve wireless in a **networked** context

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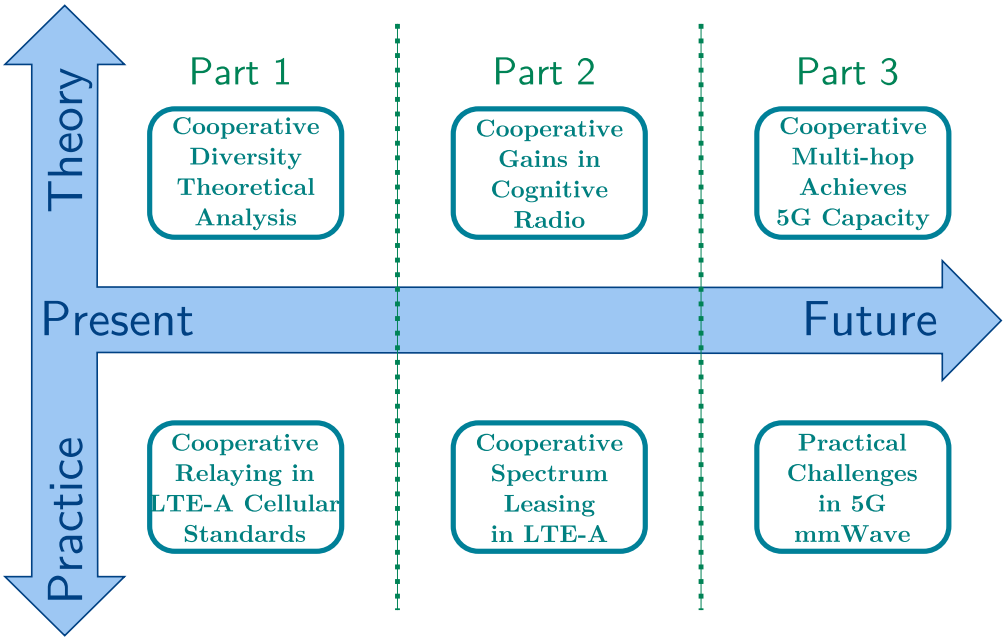


Part 1 :: Theoretical Aspects

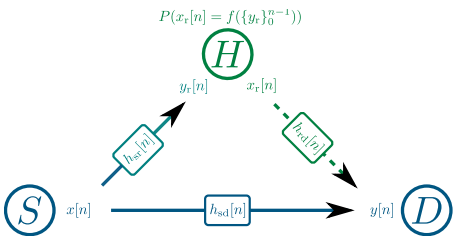
The Theory of Cooperative Communications

Cooperative Diversity for Wireless Networks

1. Information theoretic models
2. PHY layer cooperative signal
3. MAC layer functional taxonomy

Information Theoretic Model

Contribution C1.1.1: IT Survey



Relay Channel

$$
\begin{aligned}
y_r[n] &= h_{sr}[n] * x[n] + z_r[n] \\
y[n] &= h_{sd}[n] * x[n] + h_{rd}[n] * x_r[n] + z_r[n] \\
& p.d.f. P(x_r[n] = f(\{y_r\}_0^{n-1}))
\end{aligned}
$$

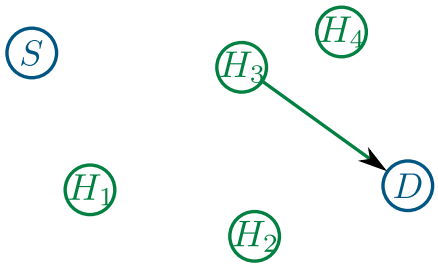
Variety of relaying techniques

1. Relay function $P(X_r = f(Y_r))$ (AF, DF...)
2. Time division (static vs dynamic)
3. Orthogonality (NBL-AF)
4. Multi-level encoding (EDDF)

Physical Layer Signaling

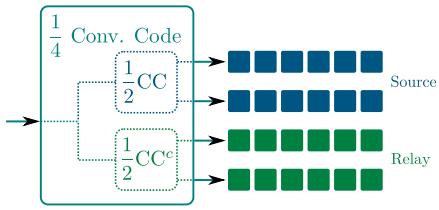
Contribution C1.1.2: PHY Survey

1 Path Selection Combining



Best Relay Cooperation

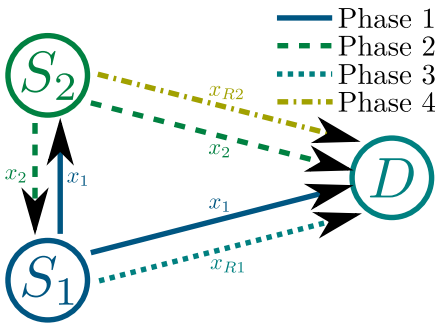
2 Channel Code Combining



Channel Coding Cooperation

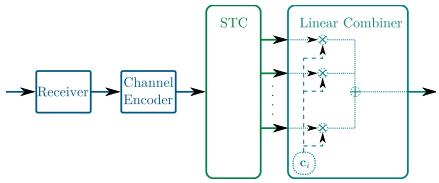
Physical Layer Signaling II

3 Network Traffic Combining



Network Coding Cooperation

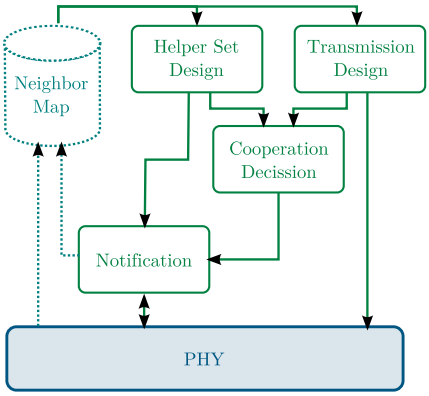
4 MIMO Combining



Space-Time Code Cooperation

MAC Layer Functions

Contribution C1.1.3: Cooperative MAC Functional Model



Functions of MAC Layer

MAC Layer Functions II

Contribution C1.1.4: MAC Survey & Classification

Neighbor Mapping

1. Possible mapped information
   1. Neighbor ID
   2. Location
   3. Willingness
   4. etc...
2. Acquisition procedure
   1. Passive or active
   2. Medium (messages, tones...)
3. Reliability
   1. Update Regularity
   2. Confirmation

Helper Set Design

1. Selected helpers from map
   1. One or many
   2. Aggregate performance metric
2. Reliability measures
   1. Default correction
   2. ACK requirement
3. Number of hops
   1. Single group between Source and Destination
   2. Multi-group-hop

MAC Layer Functions III

Cooperation Analysis and Decision

1. Compare direct and cooperative transmission
   1. Performance
   2. Cost
2. Time of decision
   1. Proactive: Evaluate before transmission
   2. Reactive: Cooperate if direct fails
3. Expected-obtained result accuracy
   1. Channels change
   2. Helpers come and go

Cooperator Notification and Agreement

1. Contact helpers and agree parameters
   1. Initialization values
   2. Measure and exchange CSI
2. Responsability of decision
   1. Source
   2. Relay
   3. Destination
   4. By parts

MAC Layer Functions IV

Cooperative Transmission Design

1. Selected helpers from Map
   1. One or many
   2. Aggregate performance metric
2. Reliability measures
   1. Default correction
   2. ACK requirement
3. Number of Hops
   1. Single group between source and destination
   2. Multi-group-hop

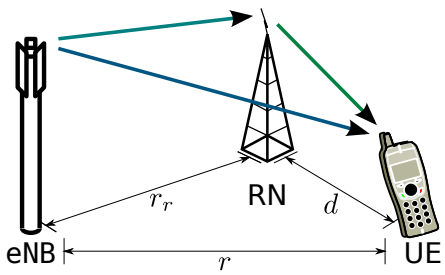
Part 1 :: Practical Aspects

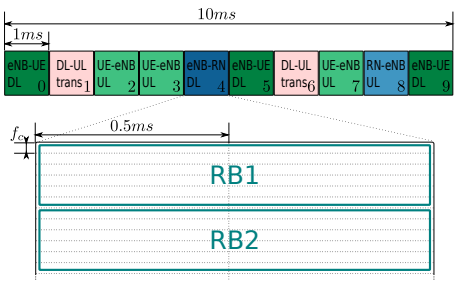
Current Cooperative Communications

Practical Multi-hop Relaying in LTE-A

1. SINR model vs System Level Simulation
2. Interference discussion
3. Scheduling analysis
4. Relay admission

SINR model vs System Level Simulation





Literature:

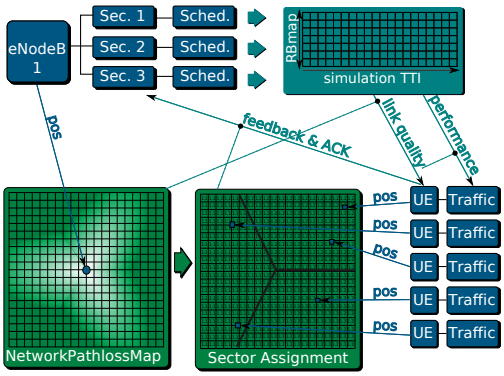
1. Single-user two hop link
2. Time division factor α
3. SINR-based rate calculations
   1. First link αR_1
   2. Second link $(1-\alpha)R_2$
4. Optimal division **per link**: balanced

$$\alpha^* = \frac{R_2}{R_1 + R_2} \quad R^* = \frac{R_1 R_2}{R_1 + R_2} .$$

Standard:

1. MAC allocates RBs
   1. Among many users
   2. In 10 static subframes
2. Rate affected by
   1. SINR
   2. RB allocation
   3. Traffic arrivals
3. Exclusive eNB-RN subframes
   1. Forced **global** α
   2. Suboptimal division

System Level Simulation: Tools

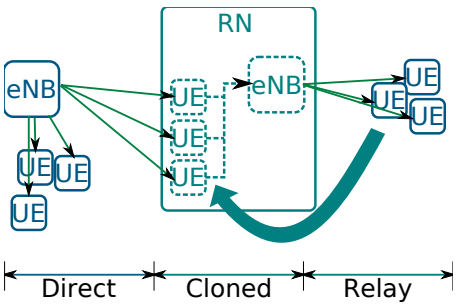


Viena LTE System Level Simulator

1. 3GPP Pathloss models for macro and femto cells
2. Radio Resource Control discrete simulation
3. Traffic generation, queuing and delivery mechanisms

System Level Simulation: Implementation

Contribution C1.2.5: Relay Simulation Extension



Relay Model

1. **Cost-effective scenario implementation**
2. Reuse simulator components
3. Plug-in scheduler API
4. Static RN frames enforcement
5. Realistic traffic forwarding

See implementation

Interference Discussion: Interference Increase in Vain

Contribution C1.2.6: Discussion of Interference and Mitigation

1. Unbalanced links
   1. Too many RN ransmissions
   2. No actual data delivery
   3. Increase interference but not rates

Default Behavior

2. Scenario with 6 DL subframes
   1. 1 eNB-RN active subframe ($\alpha = 1/7$)
   2. Default behavior: 5 RN-UE subframes
   3. Fix: add 3 blank subframes

Corrected Behavior

Scheduler Analysis I

Contribution C1.2.7: Analysis of Proportional Fair Scheduling

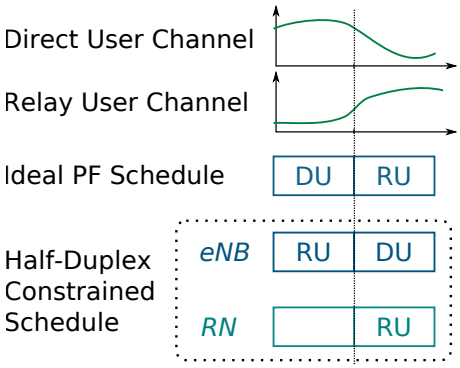
Scheduling:

1. RB allocation to users in each subframe
2. Algorithms:
   1. Round Robin
   2. Max Throughput
   3. Proportional Fair
   4. ...

Observation:

1. RNs generate rate gain with RR, but not with PF

Explanation: RNs force connections out of ideal PF slots



Static Relaying vs PF Scheduling

Scheduler Analysis II

Theoretical Analysis:

PF permanent regime ≡ Stability of ODE system

$$0 = \frac{1}{\lambda_1} - \frac{\lambda_1 \bar{\theta}_1^2}{\left(\lambda_1 \bar{\theta}_1 + \lambda_2 \bar{\theta}_2\right)^2} - \bar{\theta}_1$$

$$0 = \frac{1}{\lambda_2} - \frac{\lambda_2 \bar{\theta}_2^2}{\left(\lambda_1 \bar{\theta}_1 + \lambda_2 \bar{\theta}_2\right)^2} - \bar{\theta}_2$$

Extension to RNs:

1. Two types of connections
   1. n_d direct users
   2. n_r relay users
2. Two types of frames:
   1. α all connections allowed
   2. $1-\alpha$ only direct connections

Result: approx. time-sharing

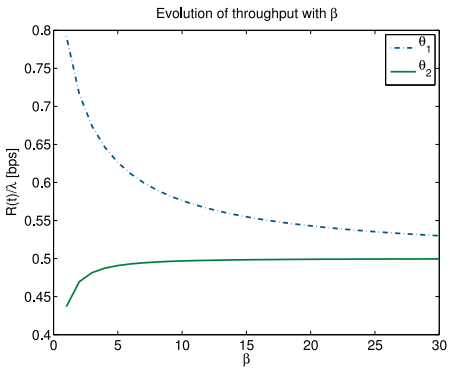
1. $\alpha \times$ rate of a $n_\mathrm{d} + n_\mathrm{r}$-user PF scheduler
2. $+(1-\alpha) \times$ rate of a n_d-user PF scheduler

See analysis

Scheduler Analysis III

Contribution C1.2.8: Scheduling Parameters Trade-offs

Incentive parameter $\beta > 1$



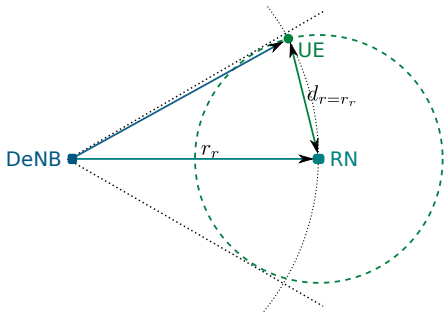
1. Proposed in literature for link balancing
2. Bias to RN selection
3. Degrades multi-user diversity

Similar analises conclude that

1. Interference $\frac{\lambda_\mathrm{d}^{(2)}}{\lambda_\mathrm{d}}$ makes direct users worse
2. Time-split factor $\alpha \to 0$ causes RN starvation
3. Time-split factor $\alpha \to 1$ causes two-hop link inbalance
4. Recommendation: increase α and make incentive $\beta < 1$ (reverse incentive)

Admission Control

Contribution C1.2.9: Relay Admission Control Algorithm



RN service area border

1. Area served by a RN
   1. Small
   2. Approx. circular
2. Estimate service distance
   1. Use approximate 3GPP pathloss for eNB-RN, eNB-UE and RN-UE
   2. Find the point that satisfies $r = r_r$
   3. Such that end-to-end rates are equal
   4. Compute radius of circumference $d_{r=r_r}$
3. Reject RNs with small or overlapping $d_{r=r_r}$

Part 2 :: Theoretical Aspects

Analysis of Cooperative Diversity in

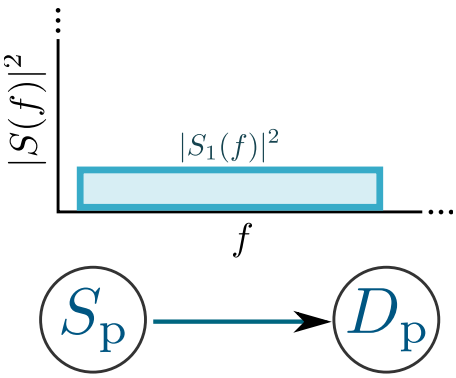
Cognitive Radio Spectrum Leasing

1. A model of Social Gain
2. Characterization of Mutual Information
3. Ergodic gain with long-term static decision
4. Distribution of gain with short-term dynamic decisions

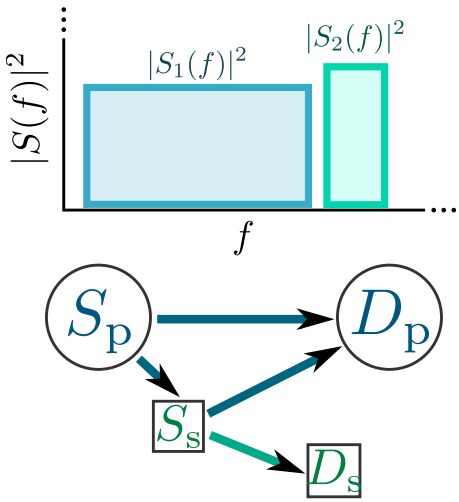
Cognitive Cooperative Model

Cooperative Spectrum Leasing: a primary spectrum licensee...

...inefficient alone,...

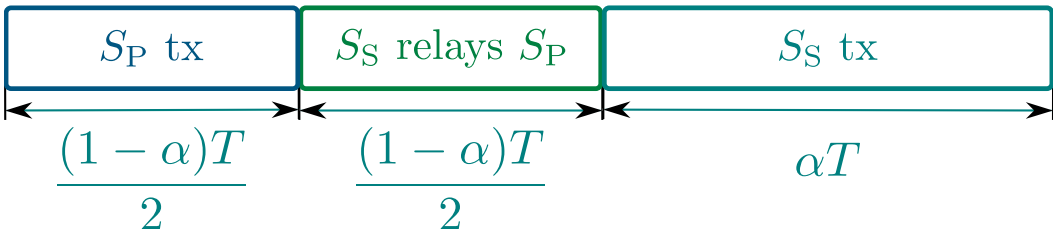


...improved with cooperation,...



... and releases part of its bandwidth to the secondary in return.

Social Gain Quantification



Resource (time) division scheme. α: fraction of resources released

Contribution C1.3.1: Metric for Social Utility

Literature

1. $\alpha = \arg\max \sum_i \mathcal{U}(R_i)$
2. *Individual* spectrum gain
3. Game-theoretic & competitive

Who gets how much?

Our model

1. $\alpha_\mathrm{max} := \alpha / R_{coop} = R_{direct}$
2. *Social* spectrum gain
3. Evaluate value of cooperation

Should we make CR like this?

Probability of Gain

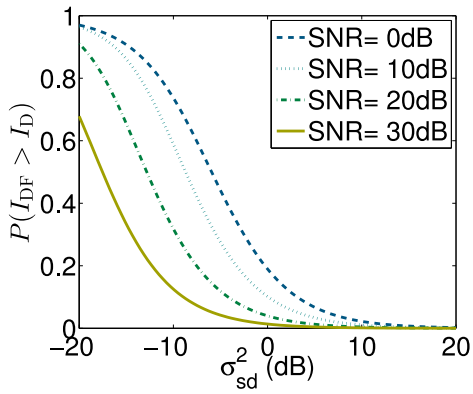
Direct transmission

$$I_{\mathrm{D}} = \log_2(1 + |h_{\mathrm{sd}}|^2 \mathrm{SNR})$$

Decode and Forward

$$I_{\mathrm{DF}} = \min \begin{array}{c} \log_2(1 + |h_{\mathrm{sd}}|^2 + |h_{\mathrm{rd}}|^2 \mathrm{SNR}) \\ \log_2(1 + |h_{\mathrm{sr}}|^2 \mathrm{SNR}) \end{array}$$

Contribution C1.3.2: Probability of Gain

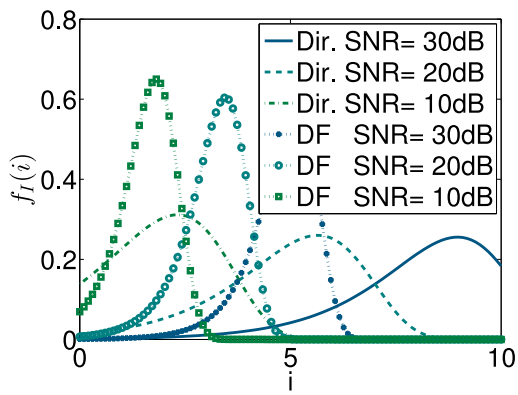


Mutual Information Distribution

1. $$f_{I_{\mathrm{D}}}(i_{\mathrm{D}}) = \frac{\ln(2)}{\mathrm{SNR}} 2^{i_{\mathrm{D}}} e^{-\lambda_{\mathrm{sd}} \frac{2^{i_{\mathrm{D}}}-1}{\mathrm{SNR}}}$$

2. $$f_{I_{\mathrm{DF}}}(x) = \begin{cases} \frac{\ln(2)}{\mathrm{SNR}} 2^{i_{\mathrm{DF}}} \left(1 + (\lambda_{\mathrm{sd}}^2 + \lambda_{\mathrm{sd}}\lambda_{\mathrm{sr}}) \frac{2^{i_{\mathrm{DF}}}-1}{\mathrm{SNR}}\right) e^{-(\lambda_{\mathrm{sd}}+\lambda_{\mathrm{sr}}) \frac{2^{i_{\mathrm{DF}}}-1}{\mathrm{SNR}}} & \lambda_{\mathrm{sd}} = \lambda_{\mathrm{rd}} \\ \frac{\ln(2)}{\mathrm{SNR}} 2^{i_{\mathrm{DF}}} \frac{\left[(\lambda_{\mathrm{sd}}\lambda_{\mathrm{rd}} - \lambda_{\mathrm{sd}}\lambda_{\mathrm{sr}}) e^{-(\lambda_{\mathrm{rd}}+\lambda_{\mathrm{sr}}) \frac{2^{i_{\mathrm{DF}}}-1}{\mathrm{SNR}}} + (\lambda_{\mathrm{rd}}\lambda_{\mathrm{sr}} - \lambda_{\mathrm{sd}}\lambda_{\mathrm{rd}}) e^{-(\lambda_{\mathrm{sd}}+\lambda_{\mathrm{sr}}) \frac{2^{i_{\mathrm{DF}}}-1}{\mathrm{SNR}}}\right]}{\lambda_{\mathrm{rd}} - \lambda_{\mathrm{sd}}} & \lambda_{\mathrm{sd}} \neq \lambda_{\mathrm{rd}} \end{cases}$$

Contribution C1.3.3: M.I. p.d.f.'s



Static Decision-Making

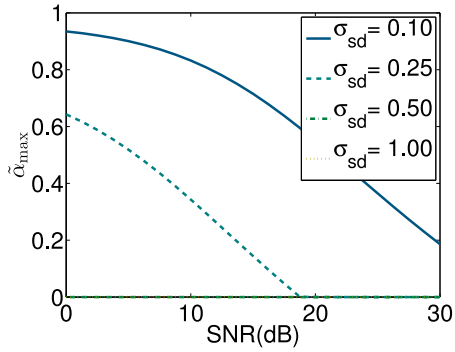
Contribution C1.3.4: Average gain with statistic CSI

1. **Only statistics of the channel available**

2. Definition of leasing limit

$$\tilde{\alpha}_{\max} = \max\left(0, 1 - \frac{E\left[I_{\mathrm{D}}\right]}{E\left[I_{\mathrm{DF}}\right]}\right)$$

3. Static decission-making: select $\alpha \in [0, \tilde{\alpha}_{\max}]$ once for all realizations



Dynamic Decision-Making

Contribution C1.3.5: Distribution of dynamic gain with instantaneous CSI

1. **Full CSI available**

2. Definition of leasing limit

$$\alpha_{\max} = \max(0, 1 - \frac{I_{\mathrm{D}}}{I_{\mathrm{C}}})$$

3. Dynamic decission-making: select $\alpha \in [0, \tilde{\alpha}_{\max}]$ for each realization

LEASING FRACTION DYNAMIC VS STATIC

σ_{sd}	SNR	$E\left[\alpha_{\max}\right]$	$\tilde{\alpha}_{\max}$
1	$0dB$	8.7%	0%
	$30dB$	0.3%	0%
0.5	$0dB$	27.0%	0%
	$30dB$	1.4%	0%
0.25	$0dB$	57.5%	64.3%
	$30dB$	5.2%	0%
0.1	$0dB$	93.1%	93.4%
	$30dB$	24.0%	18.6%

Part 2 :: Practical Aspects

Practical Cooperative Spectrum Leasing:

Third-Party Relaying in M2M LTE-A

1. Third party RN operation in LTE-A
2. Gain with a single relay
3. Gain with massive relaying
4. Case studies of interest

Third Party RN CSL Model

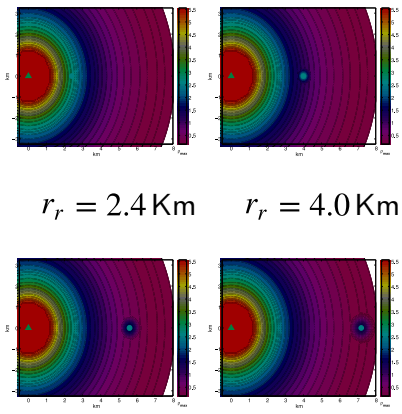
Future LTE-A link rate

1. Theoretical models from literature
2. Spectral efficiency ρ_x
3. Assigned RBs n_x
4. Throughput $T_x = \rho_x n_x$

1. Primary DeNB-UE link allocated n_{d} RBs
2. UE direct connection ρ_{d}
3. Cognitive RN offers ρ_{e2e}
4. DeNB decision
   1. Spectral efficiency gain $\Delta\rho = \max(\rho_{\mathrm{e2e}} - \rho_{\mathrm{d}}, 0)$
   2. Allocate n_{e2e} such that
   $$n_{\mathrm{e2e}}\rho_{\mathrm{e2e}} = T_{\mathrm{d}} = n_{\mathrm{d}}\rho_{\mathrm{d}}$$
   3. RB gain $\Delta n = \max(n_{\mathrm{d}} - n_{\mathrm{e2e}}, 0)$

Gain of a Single RN

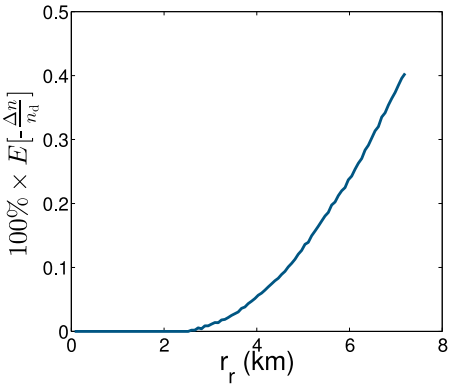
Contribution C1.4.6: Model and gain of a single RN



$r_r = 2.4\,\text{Km}$ $r_r = 4.0\,\text{Km}$

$r_r = 5.6\,\text{Km}$ $r_r = 7.2\,\text{Km}$

$\Delta\rho$ versus RN distance

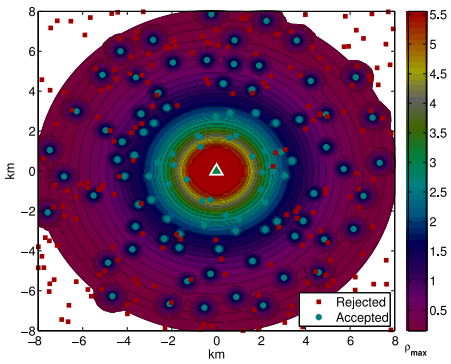


Δn versus RN distance

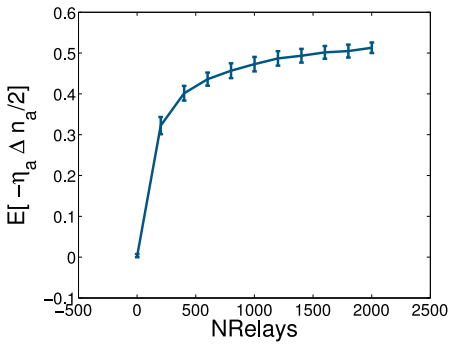
1. Very small RB gain $E\left[\Delta n\right]$ ($\Theta(0.1)\%$ RBs)
2. Coverage area very small too

Gain with Massive RNs

Contribution C1.4.7: Model and gain with massive RNs



Relay Admission Control



Δn versus no. of RNs

1. More than 100 RNs accepted (when no. candidates >300)
2. Aggregated RB gain $E\left[\Delta n\right]$ $\Theta(20)\%$ RBs

Case Studies of Interest

Contribution C1.4.8: New Application Examples and Rates

Example 1: Smart Grid Last Mile Neighborhood Area Network

1. 3rd party RN: domestic smart meter
2. DeNB location: Energy distribution AP
3. $$\rho_{\mathrm{NAN}} = \sum_{r \in \mathcal{RN}} \rho_{\mathrm{r}} \times \mathrm{E}\left[\frac{\Delta n}{n_d}\right]$$
4. ~ 5 Mbps (0.25bps/Hz, 20MHz)
5. More than enough for SG

Example 2: Machine-to-Machine Local Area Network

1. 3rd party RN: LAN hub
2. RN small cell: M2M network
3. $$\rho_{\mathrm{LAN}} = \frac{\sum_{u \in \mathcal{U}(r)} \mathrm{E}\left[\rho_a\right]}{|\mathcal{U}(r)|} \times \mathrm{E}\left[\frac{\Delta n}{n_d}\right]$$
4. ~ 10 Mbps (0.5bps/Hz, 20MHz)
5. Much higher than typical M2M (e.g. ZigBee 250Kbps)

Part 3 :: Theoretical Aspects

Cooperative Multi-hop Achieves

The Capacity Scaling of Future Cellular Networks

1. Scaling Laws model
2. Capacity scaling laws
3. Scaling laws for practical protocols
4. The importance of a good traffic model

Drastic Changes between 4G and 5G

4G Cellular

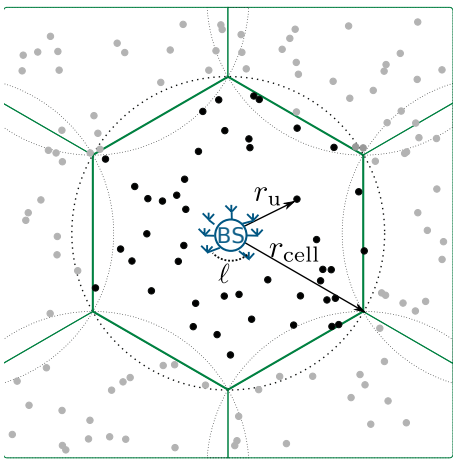
1. μWave
   1. ≤ 5GHz
   2. Scarce bandwidth
   3. Better propagation
2. $1-4$ antennas
   1. All-digital MIMO
   2. Wide-beam or omnidirectional
   3. Needs interference cancelling
3. Range ≥ 500m
   1. Lower AP density
   2. Direct link to all users
4. Wall penetration
   1. Radial cell shape

5G Cellular

1. mmWave
   1. $5-500$GHz
   2. Abundant bandwidth
   3. Worse propagation
2. 10s, 100s antennas
   1. Hybrid analog-digital MIMO
   2. Narrow directional beams
   3. Most interference directions negligible
3. Range ≤ 200m
   1. Higher AP density
   2. Multi-hop backhauling
4. Wall reflection
   1. Building-defined cell shape

Scaling Laws Model

Random Cellular Network Scaling



BS Cell with scaling dimensions

Quantity	Scaling with n	Exponent range
Area	$A = A_0 n^{\nu}$	$[0,1]$
No. of BS	$m = m_0 n^{\beta}$	$[0,1]$
No. of BS antennas	$\ell = \ell_0 n^{\gamma}$	$[0,1-\beta]$
Bandwidth	$W = W_0 n^{\psi}$	$[0,\infty]$
No. of fixed RNs	$k = k_0 n^{\rho}$	$[\beta,1]$

Scaling Laws Model

Characterization of Capacity Region

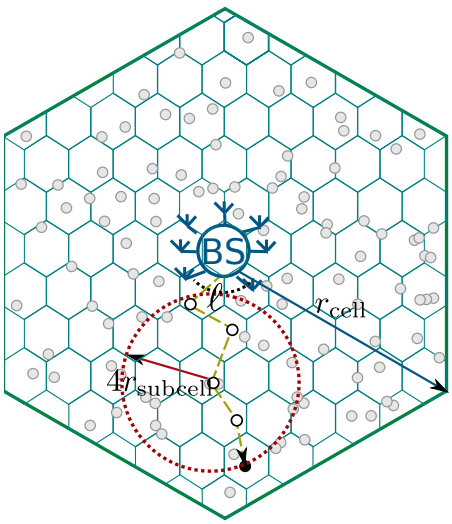
Def. DL/UL throughput $R(n)$ is *feasible* if the BSs can transmit to/receive from all nodes $R(n)$ bps at the same time

Def. The *throughput capacity scaling* of a network is $\Theta(n^x)$ iff

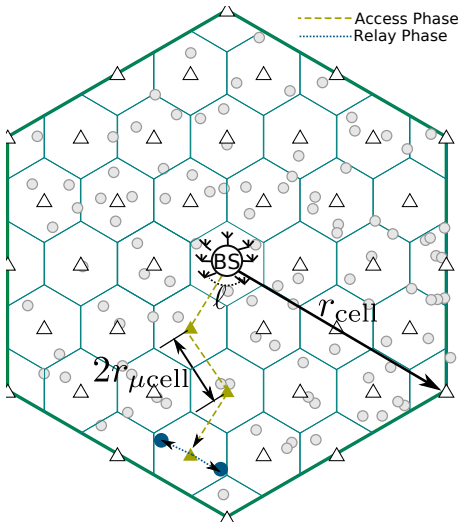
$$\exists c_1 < c_2 \Big/ \begin{array}{l} \lim_{n\to\infty} P\left(R(n) = c_1 n^x \text{ is feasible}\right) = 1 \\ \lim_{n\to\infty} P\left(R(n) = c_2 n^x \text{ is feasible}\right) < 1 \end{array}$$

Protocol Models

Multi-hop Protocols



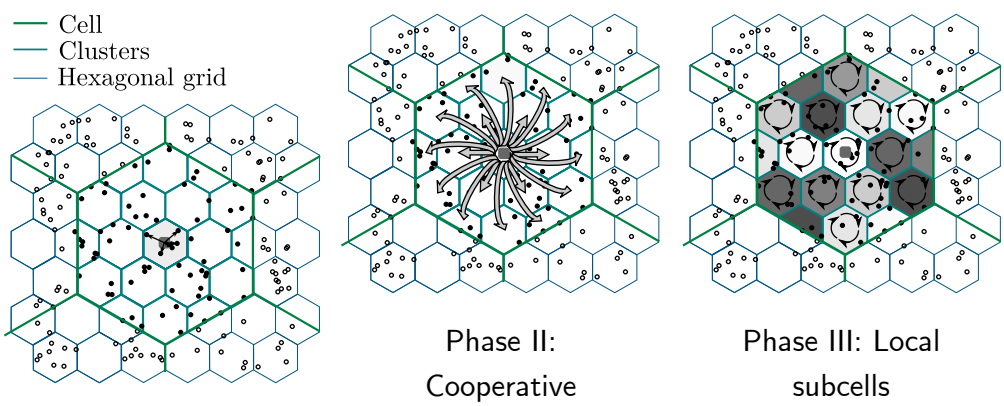
Infrastructure Multi-Hop



Infrastructure Relay-multi-Hop

Protocol Models

Infrastructure Hierarchical Cooperation Protocol

1. 

Phase I: BS subcell | Phase II: Cooperative MIMO | Phase III: Local subcells

2. Additional layers recursively employed in phases I and III

Capacity Scaling Law

Contribution C1.5.1: Throughput Capacity Scaling Laws for Wideband Cellular Networks

Th. The rate from the BS to each node is upper bounded by a function with scaling

$$R_{\mathrm{DL}(n)} \leq \Theta\left(n^{\beta-1+\min\left(\psi+\gamma,(1-\nu)\frac{\alpha}{2}\right)}\right)$$

Th. IMH DL feasible rate per node scales as

$$R_{\mathrm{IMH-DL}}(n) \sim \Theta\left(n^{\beta-1+\min\left(\psi+\gamma,(1-\nu)\frac{\alpha}{2}\right)}\right)$$

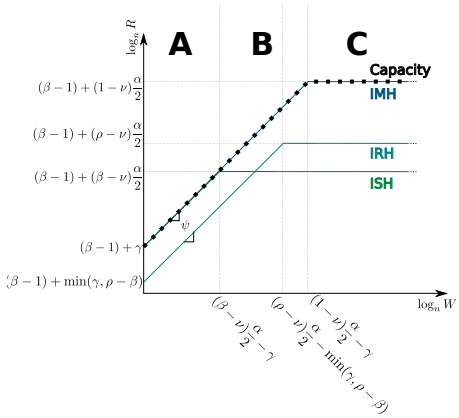
UB proof sketch

IMH proof sketch

Capacity Scaling Law

Contribution C1.5.2: Critical Bandwidth Scaling Limit

Contribution C1.5.3: Only Cooperative Multi-hop Achieves Capacity Scaling in All Regimes



1. Bandwidth Scaling Exponents
2. All links DoF limited at small bandwidth
3. As $W \to \infty$, links become power limited depending on the protocol.

Better Traffic Model

Contribution C1.5.4:Data Flow Models Are Critical

Ad-hoc traffic

1. Random destinations across the whole network
2. Infrastructure use optional
3. Direct transmission suboptimal
4. Multi-hop for extense networks
5. Hierarchical Cooperation for dense networks

Cellular traffic model

1. Nearest destination, short range
2. Must reach infrastructure
3. Direct transmission for dense narrowband network
4. Multi-hop optimal in all networks
5. Hierarchical cooperation unnecessary

Part 3 :: Practical Aspects

Challenges of Cooperative

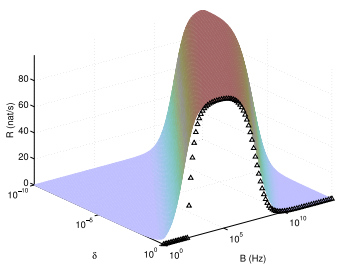
Fifth Generation Cellular Networks

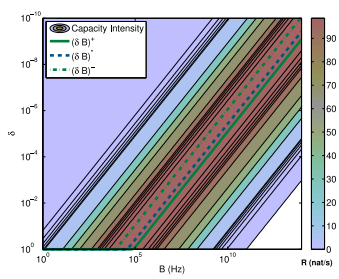
1. Intormation Theoretic Open Questions.
   1. Value of very large bandwidth
   2. Multi-user in wideband
2. Physical layer strategies.
   1. Signaling with a bandwidth limit
   2. Massive SIMO to bypass BW limitation
3. MAC layer algorithms.
   1. Resource Allocation
   2. Topology Variations

Value of Very Large Bandwidth

Contribution C1.6.5: Unified Bandwidth Occupancy Framework for Peaky and Non-peaky Wideband Signals

Capacity bounds vs BW and δ in non-coherent channel (w/ J. Du)





1. Bandwidth B
2. Peakyness factor
   $$P(|x|^2 > 0) = \delta$$
3. $SNR = \frac{P}{N_0 B \delta}$
4. Bandwidth occupancy $B\delta$

Near-linear-in-power at the critical point $(B\delta)^*$

$$\lim_{B\delta \to (B\delta)^*} \frac{C(B\delta)}{B} = N_{\mathrm{r}} \mathrm{SNR} - \frac{N_{\mathrm{r}}(N_{\mathrm{r}} + N_{\mathrm{t}})}{N_{\mathrm{t}}^2} \mathrm{SNR}^{1+\alpha} + o(B^{-(1+\alpha)})$$

Multi-user in Wideband

Contribution C1.6.6: Frequency Division Achieves Critical Bandwidth - Analysis of Multiuser Channels

Def. The *multi-user critical bandwidth* (B_x^{crit}) of a multi-user channel x is the bandwidth above which the rate region does not grow.

Analysis of N-user MAC

1. Using FDMA: N Orthogonal point-to-point subchannels
2. $B_{\text{FDMA}}^{\text{crit}} = \sum_{i=1}^{N} B_i^{\text{crit}}$
3. $\Sigma R_{\text{FDMA}}^{*} = \sum_{i=1}^{N} R(B_i^{\text{crit}})$

1. Using SIC: Sum-rate limited by a $N\times$ antennas link
2. $B_{\text{SIC}}^{\text{crit}} < B_{\text{SIC}}^{\text{crit}}$
3. $\Sigma R_{\text{SIC}}^{*} < \sum_{i=1}^{N} R(B_i^{\text{crit}})$

Discussion of BC duality

Signaling with a bandwidth limit

Coherent receiver on noncoherent channel

1. Detect and enforce critical bandwidth limitation
2. Use reference-assisted channel estimation
3. Non-ideal rate degradations
   1. Dedicated pilot/data resources
   2. Imperfect estimation
   3. Time variation
   4. Necessity of feedback

Noncoherent receivers

1. Poor spectral efficiency
   1. On/off signaling
   2. Pulse Position Modulation (PPM)
   3. Frequency Shift Keying (FSK)
2. Multitone-FSK
   1. No. of active tones Q
   2. Still bandwidth limited
3. *peaky* m-FSK
   1. Duty cycle δ
   2. Still bandwidth occupancy limited

Massive SIMO to bypass BW limitation

Contribution C1.6.7: Massive MIMO Mitigation of Overspreading in Wideband Channels

Wideband analysis

1. Fixed receive antennas N_r
2. $B \to \infty \Rightarrow SNR \to 0$
3. Rate nearly linear in power
4. Coherent channel :

$$\lim_{B\to\infty} B \log\left(1 + \frac{PN_r}{N_0 B}\right) = \Theta\left(\frac{PN_r}{N_0}\right)$$

5. Non-coherent channel:

$$\Theta\left(N_r \mathrm{SNR} - \frac{N_r(N_r + N_t)}{N_t^2}\,\mathrm{SNR}^{1+\alpha}\right)$$

Massive MIMO analysis

1. Fixed bandwidth B
2. $N_r \to \infty \Rightarrow SNR \to \infty$
3. Rate linear in bandwidth
4. Coherent channel :

$$\lim_{N_r\to\infty} B \log\left(1 + \frac{PN_r}{N_0 B}\right) = \Theta\left(W \log(N_r)\right)$$

5. Non-coherent channel: No *combining gain*

Massive SIMO to bypass BW limitation II

Joint analysis (w/ M. Chowdhury and A.Manolakos)

1. $B \to \infty$ and $N_r \to \infty$, ¿SNR?
2. $\epsilon := \lim_{B,N_r\to\infty} \frac{\log B}{\log N_r}$
3. $B = N_r^{\epsilon}$
4. Rate limited by

$$R = \begin{cases} \Theta(W) & \epsilon \le \epsilon_{th} \\ \Theta(\frac{P}{N_0}) & \epsilon > \epsilon_{th} \end{cases}$$

5. Coherent channel trivial
   1. Coherent combining gain N_r
   2. $\mathrm{SNR} \propto \frac{N_r}{B}$
   3. $\epsilon < 1 \Rightarrow \mathrm{SNR} \to \infty$
   4. $\epsilon > 1 \Rightarrow \mathrm{SNR} \to 0$

Non coherent channel ϵ_{th}

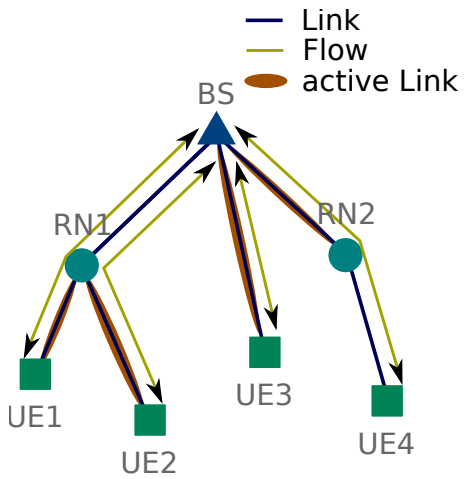
1. Critical Bandwidth $\propto \frac{P}{N_0}\sqrt{N_r}$, suggests $\epsilon_{th} = \frac{1}{2}$
2. But the wideband analysis does not hold when $\epsilon < 1$
3. Alternate upper bound:
   1. Received signal vector $\mathbf{y} \in \mathbb{C}^{1\times N_r}$
   2. $|\mathbf{y}|^2$ sufficient statistic
   3. $\mathrm{E}[|\mathbf{y}|^2] = \frac{P\sqrt{N_r}}{N_0 B}$
   4. If $\epsilon > \frac{1}{2}$ and $R > \Theta(\frac{P}{N_0})$, errors
4. Achievable with multicarrier Pulse Energy Modulation
5. $\sqrt{N_r}$ *diversity gain*

New mmWave cellular paradigm

LTE	mmWave
1 Single-hop	1 Multi-hop
2 Omnidirectional	2 High directivity
3 Long range	3 Shorter range
4 High BS density	4 Ultra-high BS density
5 Static global TDD/FDD	5 Dynamic TDD

Resource Allocation

Contribution C1.6.7: Dynamic Resource Allocation in a Tree-topology



Tree topology with simultaneous link activation

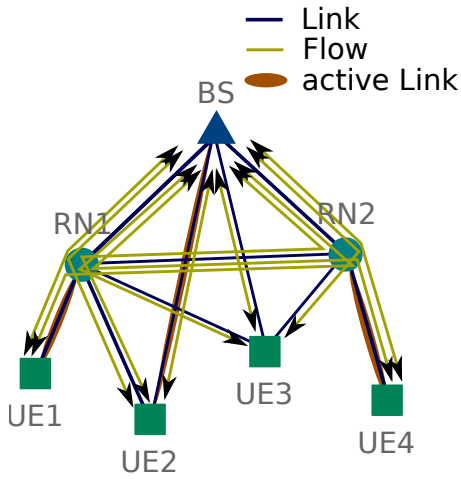
DTDD resource allocation

(w/ R.Ford)

1. **Simplification: tree topology**
2. Multi-user diversity: simultaneous connections per device
3. Binary mixed-integer non-convex optimization
4. **Closer to practical implementation**

Resource Allocation

Contribution C1.6.7: Dynamic Link Scheduling in an Arbitrary-Topology



Mesh topology with one2one link activation

DTDD link scheduling

(w/ J. García)

1. Full mesh topology
2. **Simplification: one connection per devide**
3. Max Weighted Queue Pressure (MQP)
4. Network Utility Maximization (NUM)
5. **Higher theoretical capacity**

Conclusions and Future Work

1. The point-to-point link model is nearly exhausted
2. Future wireless networks require cooperative network solutions
3. Cooperative diversity gain analysis
   1. Relay channel
   2. PHY techniques
   3. Cooperative MAC functions
4. Current LTE-A standard too conservative
   1. Fixed relaying only
   2. Insufficient understanding of multi-hop
   3. Prioritizes backwards compatibility and simplicity over optimized operation

5 Short term: cooperation through Cognitive Radio

1. Stubborn licensees use spectrum inefficiently
2. Cooperative Spectrum Leasing
3. Static scheme: only for very bad primaries
4. Dynamic scheme: minor to high gain depending on efficiency of primary

6 Potential for CSL in LTE-A

1. Puny gain with a single RN
2. Massive RN populations attractive, but still moderate gain
3. Niche applications: M2M, SG, P2P

7 Long term: next standard generation

1. Massive degrees of freedom (spatial, bandwidth, antennas...)
2. Approaching power limited capacity regimes
3. Imperative need of cooperative multi-hop
4. The less cooperative a protocol, the more limited

8 Re-think wireless networking: cooperative multi-hop by design

1. Understand the triple interaction:
   - Large bandwidth
   - Multi user allocation
   - Capacity
2. Implement massive bandwidth signaling
   - Non-peaky and coherent (pilots)
   - Peaky and non-coherent (FSK)
3. Massive no. of reception antennas
4. Dynamic Duplex link selection
   - Multi-user tree-topology practical power allocation
   - Single-user mesh-topology capacity-approaching scheduling

Research output

1. Four published journal papers
2. Plus two submitted and one nearly ready for submission
3. Six published conference papers
4. Plus two accepted for this summer
5. One research-license simulation software released

View References

THANK YOU